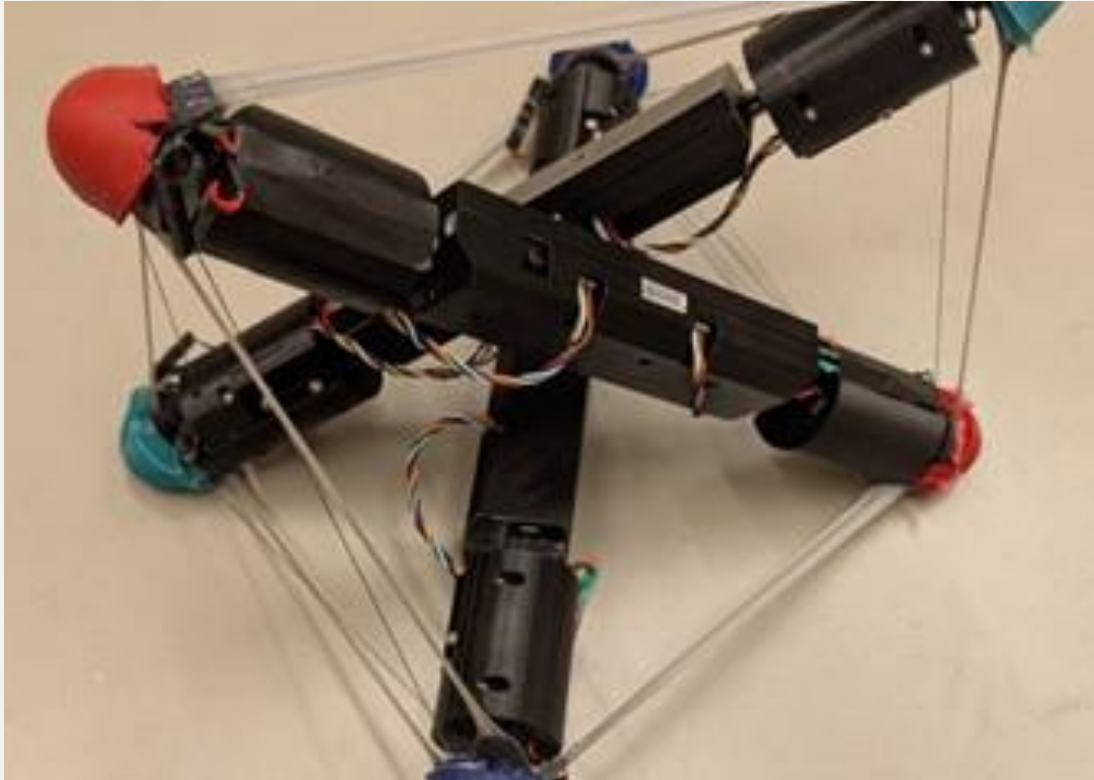


Tensegrity Robot



THREE-BAR TENSEGRITY

Yale Tribar — Johnson et al., Nat. Mach. Intell. (2026)

The mechanism is easy to see



SIX-BAR TENSEGRITY

NASA SUPERball — Sabelhaus et al., ICRA (2015)

More support faces and control choices

One motion principle

02

Yale Tribar — Johnson et al., Nat. Mach. Intell. (2026)



- 1** Selected cables become shorter
- 2** The body changes shape
- 3** The center of mass leaves the support area
- 4** Gravity tips the robot onto a new face

Both robots move by changing shape before they roll.

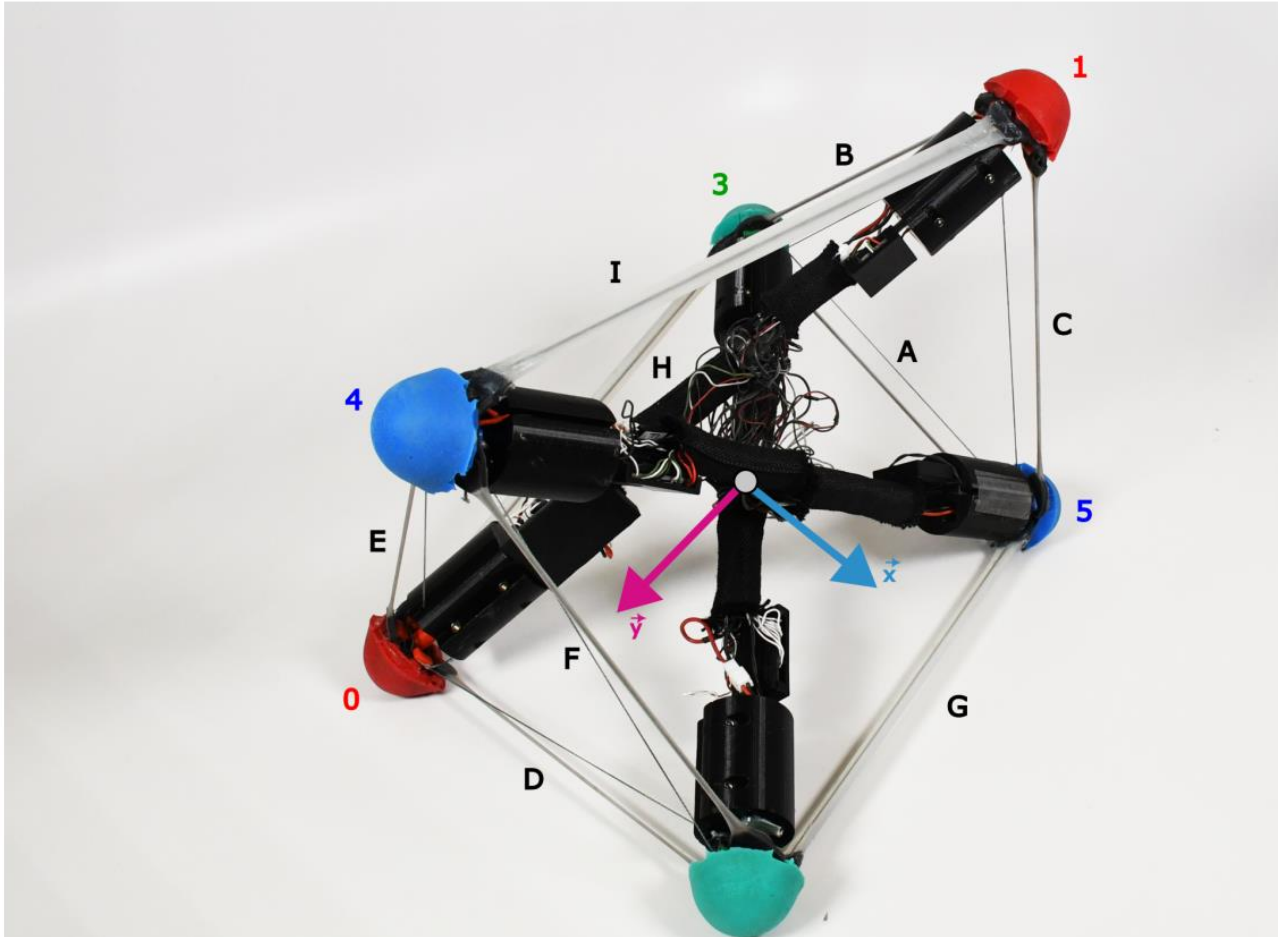
[Full three-bar gait video](#)

Three-bar structure

03

THREE-BAR

Yale Tribar — Johnson et al., Nat. Mach. Intell. (2026)



3 rigid struts
carry compression

6 active tendons
motor driven

3 passive tendons
elastic and prestressed

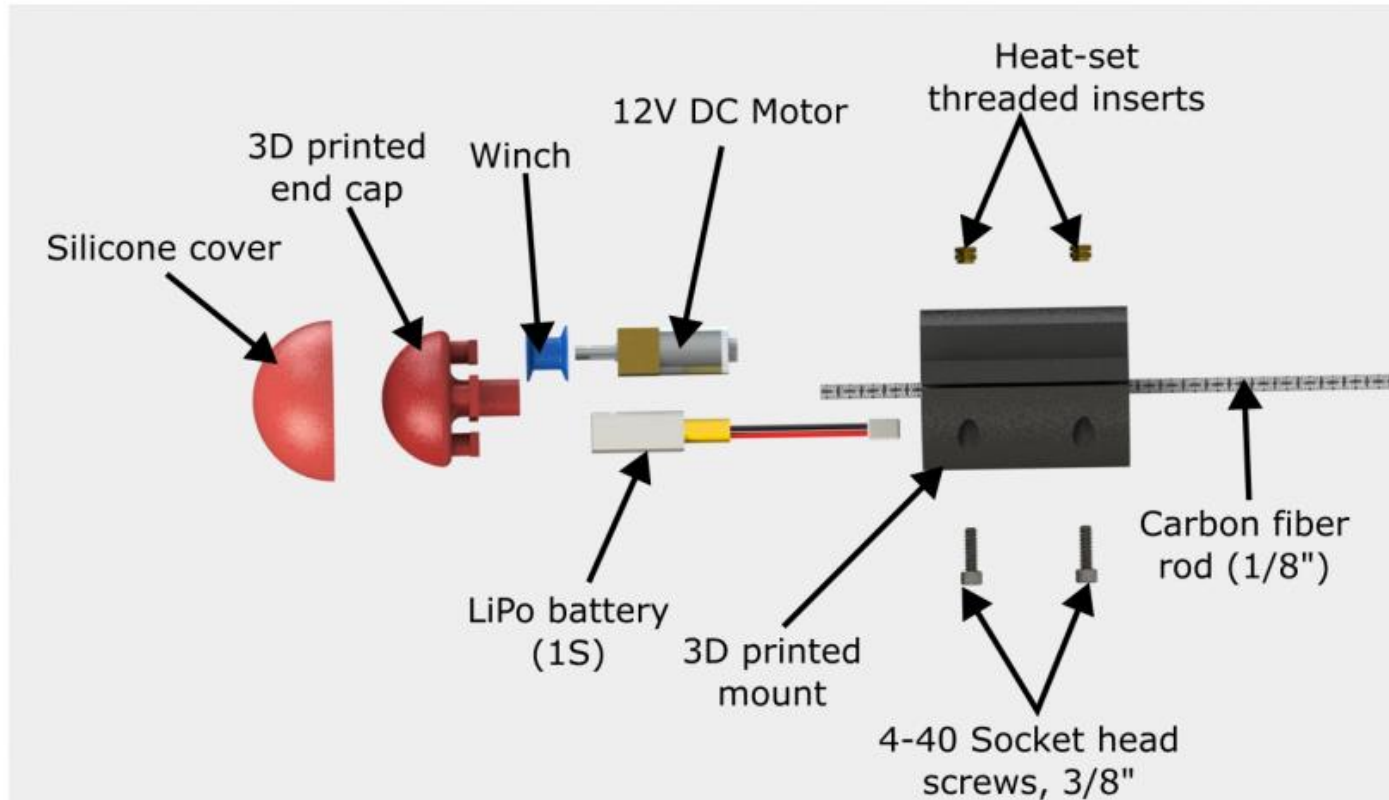
The struts do not touch each other.

One three-bar actuation design

04

YALE TRIBAR EXAMPLE

Yale Tribar — Johnson et al., Nat. Mach. Intell. (2026)



6 winches total

Two ends on each of three struts

Wind in

Shorten the tendon and pull two nodes together

Let cable out

Allow prestress to lengthen the tendon

This is one possible layout: six winches change tendon lengths.

[Yale mechanical details](#)

One quasi-static face change

05

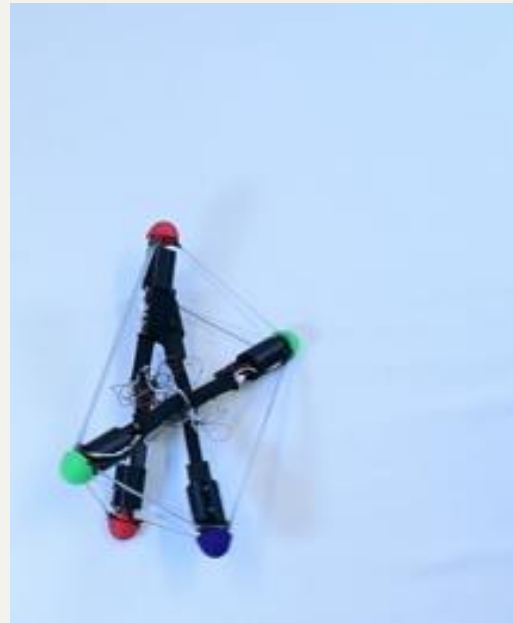
THREE-BAR MOTION PRIMITIVE

Yale Tribar — Johnson et al., Nat. Mach. Intell. (2026)



1 SUPPORT

Three end caps form a support triangle



2 DEFORM

Selected tendons shorten slowly



3 CROSS EDGE

The center of mass moves beyond one edge



4 SETTLE

Gravity brings a new face to the floor

Quasi-static describes the slow deformation phase. Tipping and landing are still dynamic.

Continuous rolling and steering

06

DYNAMIC THREE-BAR GAIT

Yale Tribar — Johnson et al., Nat. Mach. Intell. (2026)



DYNAMIC ROLLING

The next contraction starts before the body fully settles



STEERING

Asymmetric cable timing redirects the next roll

A gait repeats face changes. Timing determines how much momentum carries into the next one.

Why the three-bar robot can absorb a fall

07

Yale Tribar drop test — Johnson et al., Nat. Mach. Intell. (2026)



5.7 m

demonstrated drop height

Before impact

Return to a shape with room to deform

During impact

Elastic tendons stretch and lengthen the stopping time

This result belongs to the tested Yale robot. It does not apply to every three-bar build.

[Full drop video](#)

What changes with six bars

08

Yale Tribar — Johnson et al. (2026) • NASA SUPERball — Sabelhaus et al., ICRA (2015)

THREE-BAR

3 struts

9 structural tendons

6 active winches

Few support states make the basic roll easy to see.

SIX-BAR SUPERBALL

6 struts

24 structural cables

12 active motors on the physical robot

More support faces give the controller more choices.

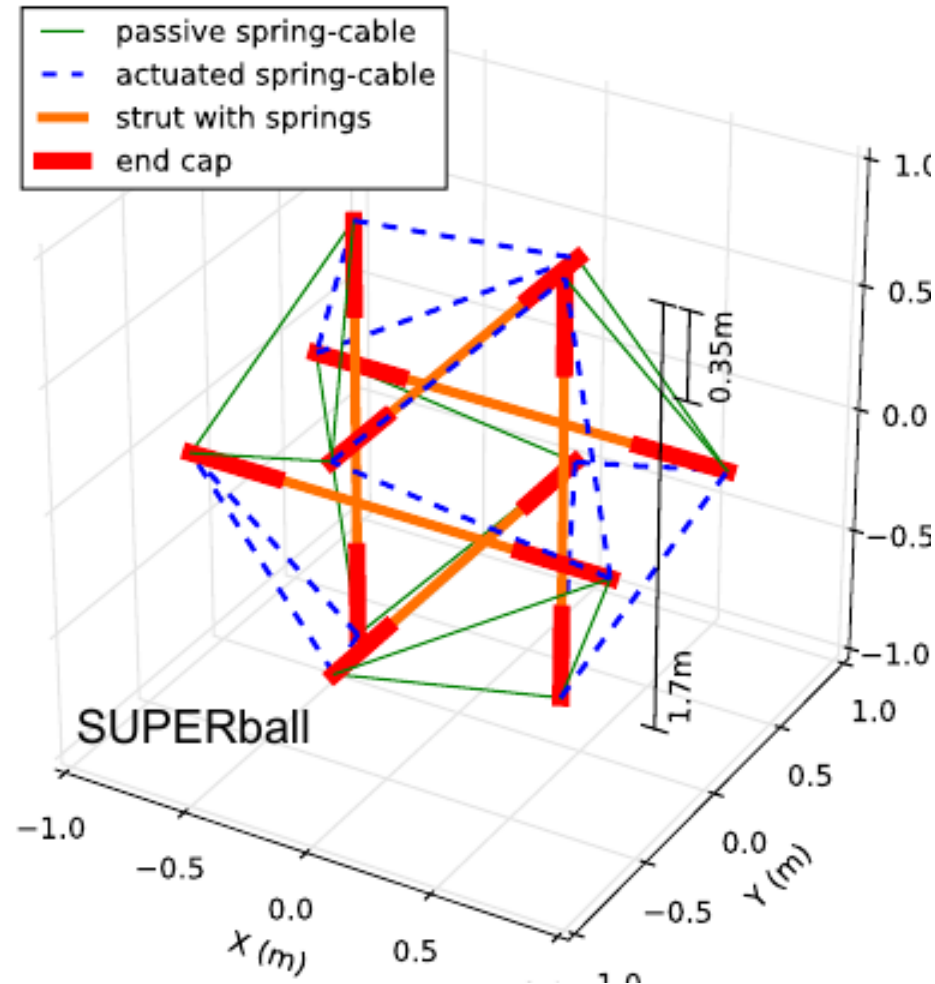
The six-bar robot uses the same tipping idea, but it needs better face identification and cable mapping.

Six-bar structure

09

SIX-BAR SUPERBALL

NASA SUPERball — Sabelhaus et al., ICRA (2015)



6 struts

12 active cables

12 passive cables

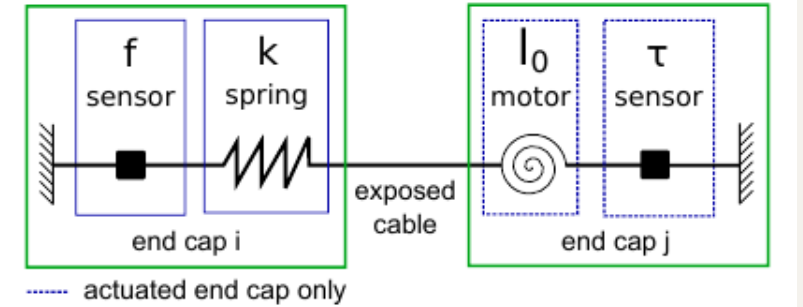
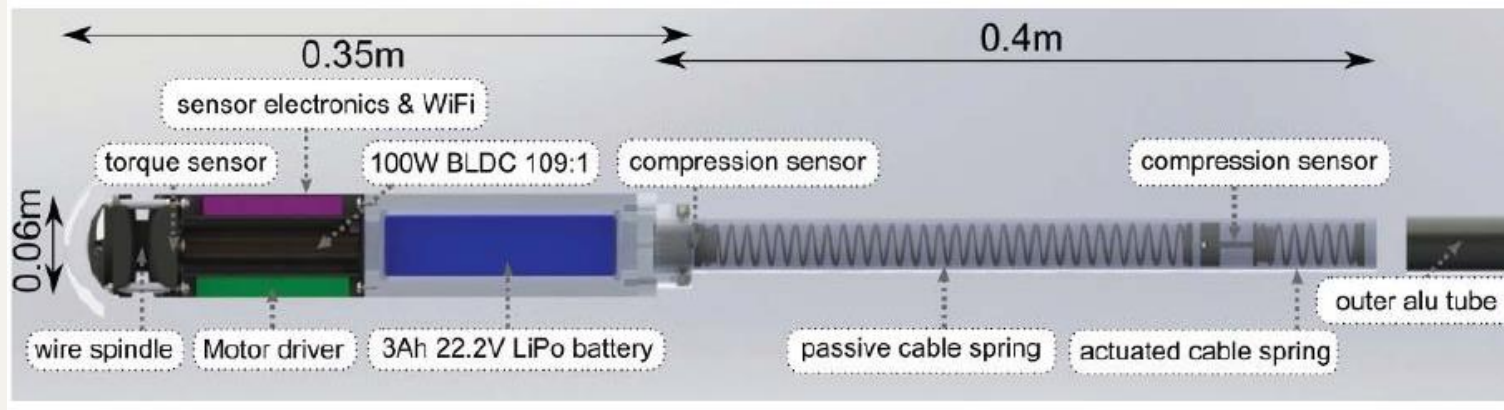
The 2015 physical robot actuates half of its structural cables.

Six-bar actuation

10

SIX-BAR SUPERBALL

NASA SUPERball hardware — Sabelhaus et al., WCSCM (2014)



12 end caps

One motor in each end cap

The spool changes one active cable's rest length.

Internal springs

They keep the cable compliant and under tension

The whole cable network redistributes the force.

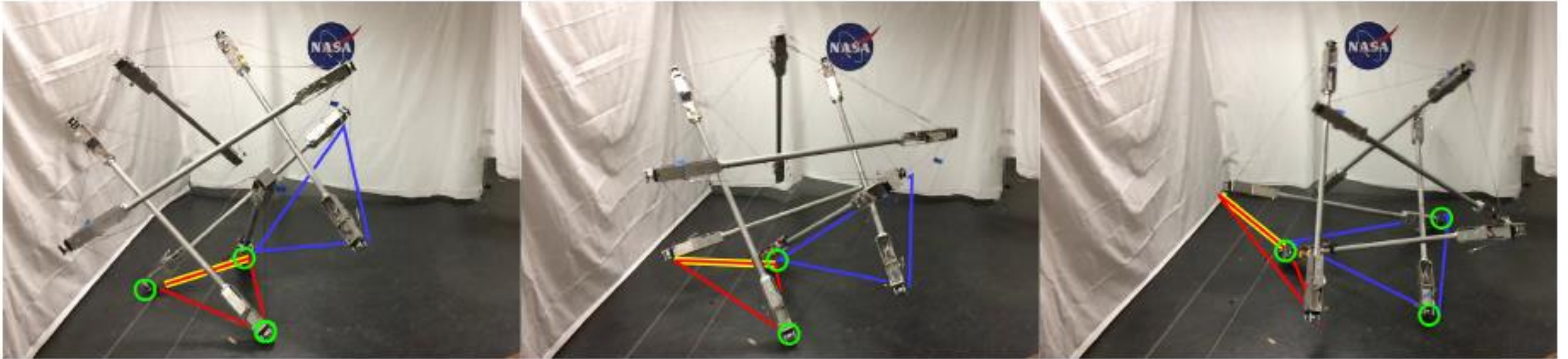
A motor commands a local cable length. The resulting body shape depends on the whole robot.

One six-bar face change

11

SIX-BAR SUPERBALL

NASA SUPERball — Sabelhaus et al., ICRA (2015)



- 1 Choose an exit edge**
- 2 Retract one bottom cable**
- 3 Let gravity complete the tip**

The red triangle is the current support face. The blue triangle becomes the new base.

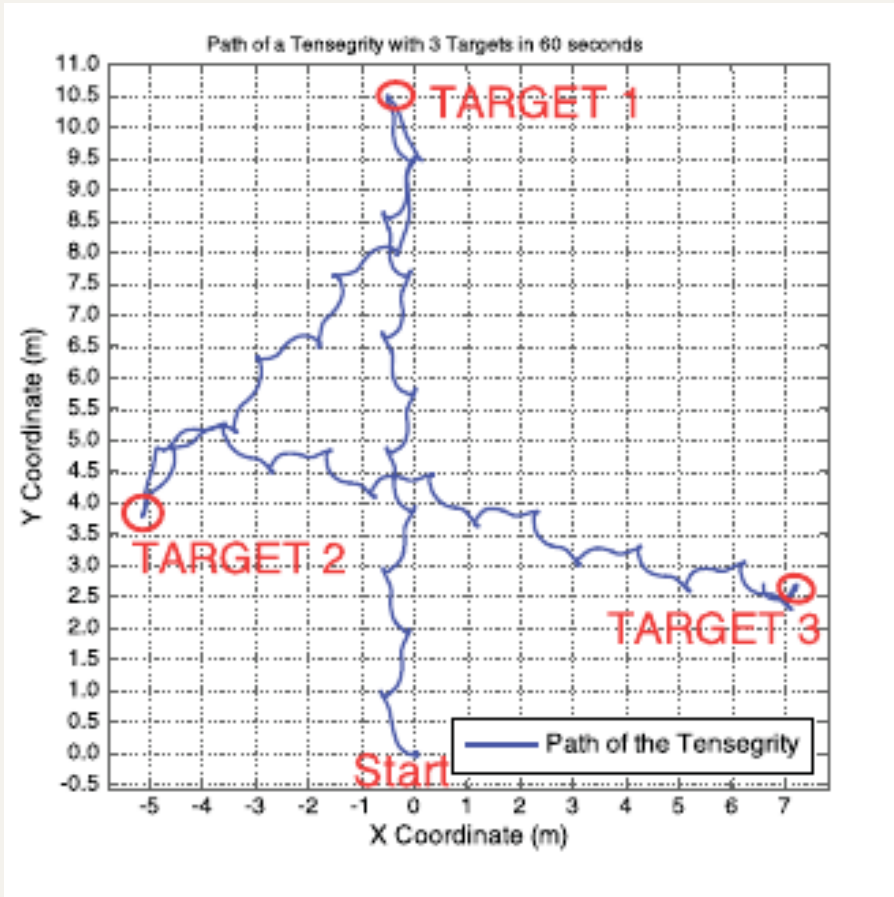
The 2015 experiment used roughly half the starting length of one cable for a single face change.

Six-bar steering

12

SIX-BAR SUPERBALL

NASA SUPERball platform — Bruce et al., ICRA (2014)



Center-of-mass path in simulation

At each landing

1. Find the current support triangle
2. Choose the edge closest to the target direction
3. Run the face-change action for that edge

The path zig-zags because each step crosses a triangular face edge.

Tensegrity locomotion in one sentence

Cable length changes move the center of mass across a support edge.

Three-bar

makes one motion cycle easy to see

Six-bar

adds more support faces and control choices

NASA SUPERball — Sabelhaus et al., ICRA (2015)

[Reproduce the mechanics in NASA NTRTsim](#)



Your project starts with two linked tasks

14

DESIGN ASSIGNMENT

Understand the mechanism → freeze a design → freeze a purchase plan

TASK 1

Design a robot

- choose three-bar or six-bar
- label every strut and tendon
- define one face-changing motion
- place the winches, electronics, and battery

TASK 2

Develop a buildable plan

- select motor, controller, and battery
- resolve every connector and harness
- submit BOM and wiring diagram for review
- plan roughly \$400–600; flag substitutions

Pre-order gate: geometry + actuator map + motor calculation + wiring path + current checkout prices

First decision: three-bar or six-bar?

15

CHOOSE THE FIRST MORPHOLOGY

Reference platforms: Yale Tribar — Johnson et al. (2026) • NASA SUPERball — Sabelhaus et al. (2015)

THREE-BAR

Recommended first build

3 struts
6 active + 3 passive tendons
6 motors on 3 active struts

Plan around \$400–600

SIX-BAR PROTOTYPE

Advanced build

6 struts
24 total = 6 active + 18 passive
6 motors on the same 3 active struts

Separate BOM required

Reusable

Three active strut modules, winches, wiring, IDs, calibration, and safety logic

Must be redesigned

Six-bar node geometry, 24-cable map, lengths, support-face logic, and mass distribution

The modules transfer. The topology does not.

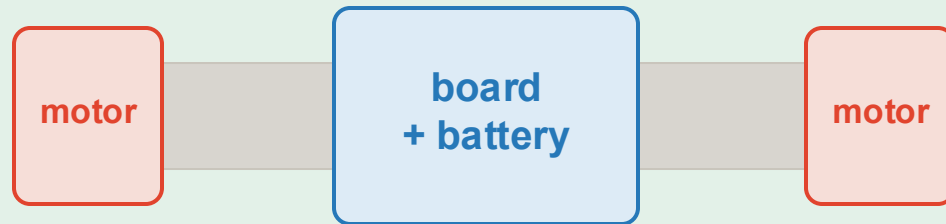
Untethered hardware can be packaged two ways

16

ARCHITECTURE CHOICE

DISTRIBUTED / ROD-CENTERED

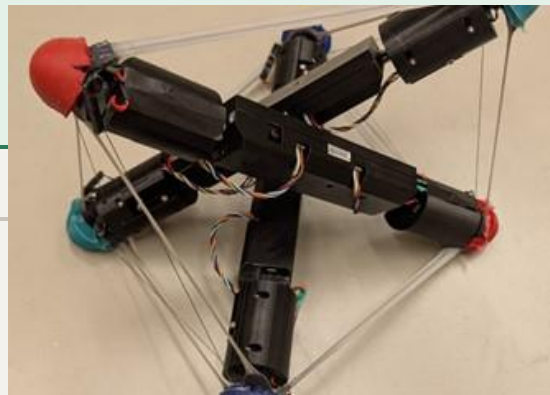
Recommended here



Short local wiring; modules are reusable.
Examples: Yale Tribar and Berkeley TT-3.

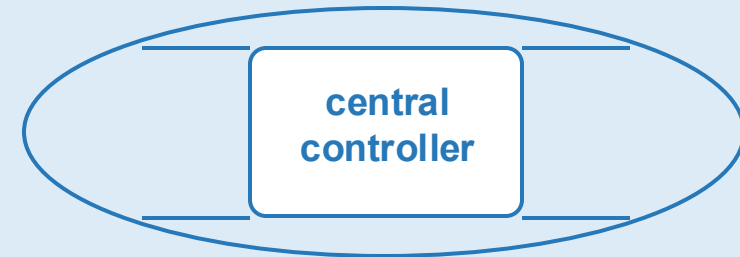
[Yale Tribar — Johnson et al. \(2026\)](#)

[Berkeley TT-3 — Chen et al. \(2017\)](#)



SUSPENDED CENTRAL CONTROLLER / PAYLOAD

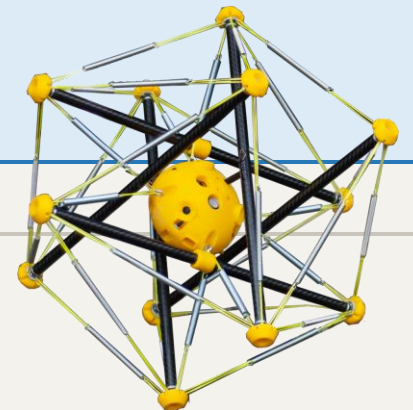
A valid alternative



Central mass is protected, but suspension, inertia, and moving wire routes become design variables. Actuators may still be distributed.

[Berkeley TT-4mini — Chen et al. \(2017\)](#)

[Central payload — Zhang et al., IASS \(2018\)](#)

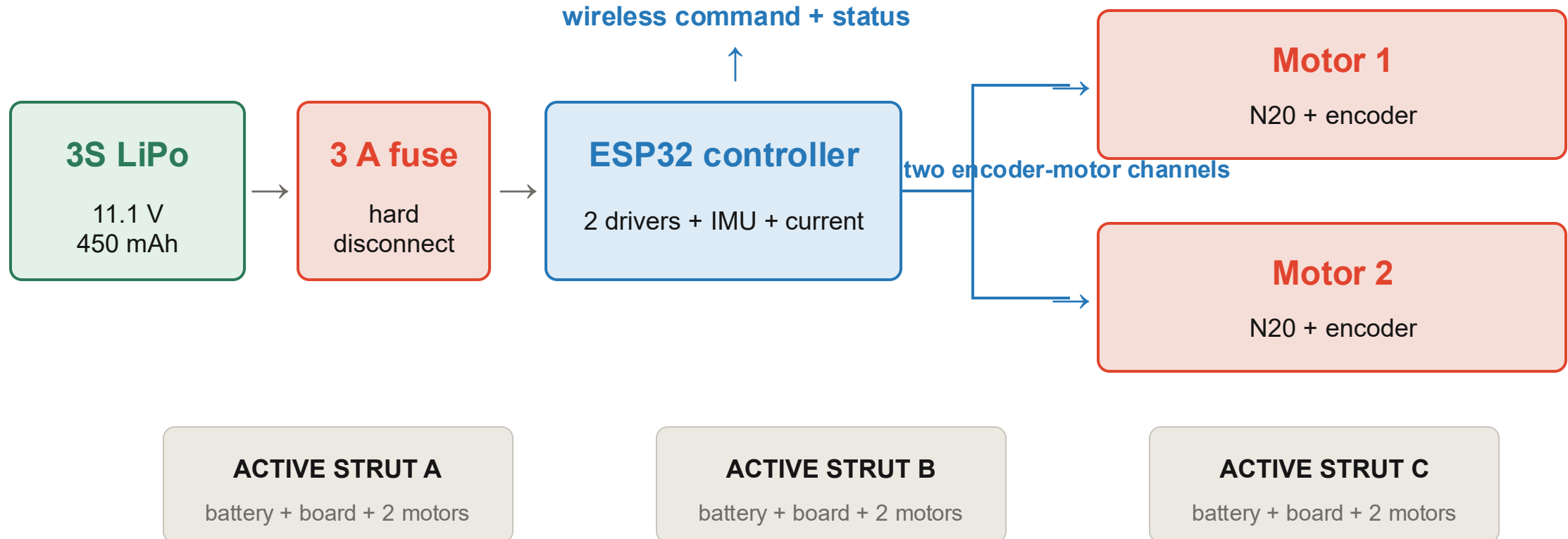


Untethered means the complete loop is onboard

17

ONE ACTIVE STRUT MODULE

Rod-centered precedents: Yale Tribar — Johnson et al. (2026) • Berkeley TT-3 — Chen et al. (2017)



During motion: no USB cable, no bench supply, and no electrical wire crossing the tendon network.

Choose a motor from the tendon requirement

18

MOTOR SELECTION

Force first

$$T_{\text{required}} = F_{\text{tendon}} \times r_{\text{spool}} \div \eta \times SF$$

Larger spool → more line speed, less cable force

Smaller spool → more cable force, less line speed

$$v_{\text{line}} = 2\pi \times r_{\text{spool}} \times \text{RPM} \div 60$$

Then check the whole system

- 1 Working torque — not stall torque
- 2 Loaded speed and required stroke
- 3 Motor and stall current
- 4 Voltage and driver compatibility
- 5 Encoder, mass, size, and gearbox life

Stall torque is a damage limit, not a normal operating point.

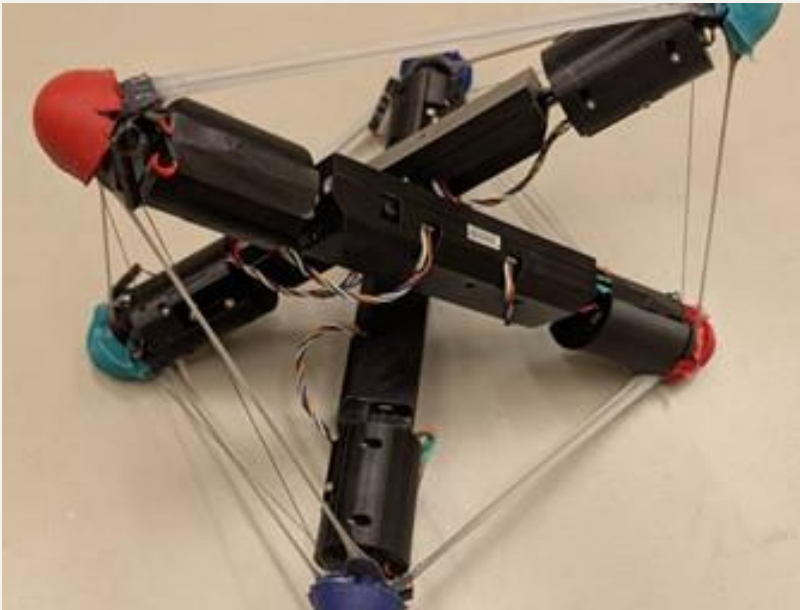
The motor, spool, driver, wire, fuse, and battery must be sized as one chain.

[DC gearmotor guidance](#)

Yale Tribar: published reference architecture

PUBLISHED OPEN-SOURCE REFERENCE

Yale Tribar hardware; Johnson et al., IEEE RA-L (2026)



3 struts · 6 motors · untethered

Each strut is a self-contained smart module.

Electronics	Per strut	Robot
Custom motherboard	1	3
Arduino Nano 33 IoT	1	3
SparkFun dual TB6612FNG	1	3
MPR121 tendon interface	1	3
Pimoroni ICM20948 IMU	1	3
Adafruit INA260 monitor	1	3
Pololu #5192, 6 V, 150:1	2	6
220 mAh LiPo cells	4	12

Open source does not mean plug-and-play

Exact replication still needs a custom PCB, assembly, strain sensors, printing, crimping, and soldering.

Yale Tribar: what one complete robot contains

SYSTEM QUANTITIES — USE THE OFFICIAL SPREADSHEET
FOR EXACT ORDER QUANTITIES

Yale official BOM and assembly guide (accessed 7 Oct 2026)

STRUCTURE + TENDONS

Published item	Qty / robot
Aluminum strut, 1/8 in × 12 in	3
Motor enclosure A + B	6 + 6
Electronics enclosure A + B	3 + 3
End cap + compliant cover	6 + 6
Cable guides + socks	6 + 6
Motor winches / set screws / spacers	6 each
Active tendons + short sensors	6
Passive tendons / long sensors	3

BOARD + ASSEMBLY HARDWARE

Published item	Qty / robot
Custom motherboards	3
Motor connectors / battery connectors	6 / 12
Charge connectors / switches	3 / 3
Sensor headers / board headers	6 / sets
MOSFETs + resistors	9 + 9
Motor jumpers + power jumpers	6 + 6
4-40 inserts + screws	30 + 30
2-56 inserts + screws	12 + 12

PUBLISHED COST REFERENCES

\$223.38 / bar ≈ \$670 / robot \$1,021.77 / 2 robots ≈ \$511 / robot + \$175.93 shared supplies

[Open the authoritative Yale BOM](#)

[Assembly guide](#)

[Mechanical details](#)

Costs are references, not a delivered quote; custom PCB, printing, fabrication, tax, and shipping vary.

Two motor options

21

BOTH ARE N20-SIZE MOTORS; THE EXACT ELECTRICAL DESIGN IS DIFFERENT

Motor data: Pololu #5218; Waveshare DCGM-N20; Toshiba TB6612FNG

SUGGESTED BASELINE

Pololu #5218

\$32.45 each · \$29.85 at 5+

**12 V | 220 rpm | 0.75 A stall
1.8 kg·cm stall torque | 11 g
JST-SH 6-pin encoder connector**

Why it is preferred

0.75 A stall is below the TB6612's conservative 1.0 A/channel operating limit. It costs more, but gives cleaner driver margin.

LOWER-COST EXPERIMENTAL OPTION

Waveshare N20

\$6.99 each

**12 V | 200 rpm | ≤ 1.1 A locked
 ≥ 1.2 kg·cm locked torque | 18 g
JST-ZH 1.5 mm 6-pin connector**

Why it is conditional

1.1 A locked current is slightly above the driver's 1.0 A/channel operating limit. Validate one active strut before buying six.

Yale reference: Pololu #5192 is 6 V. It is not either 12 V option.

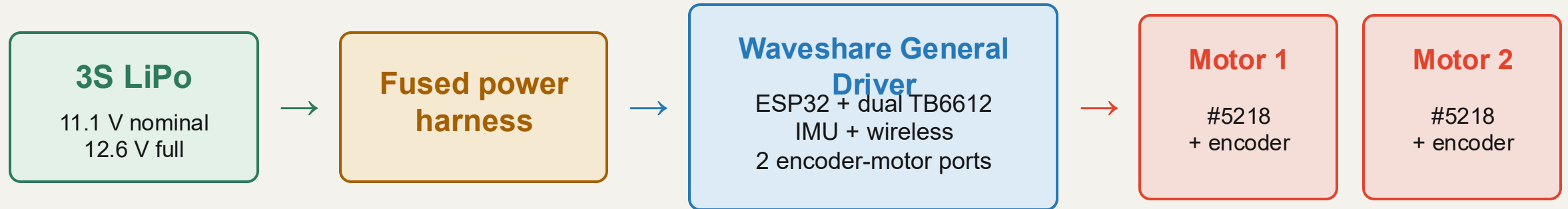
Neither option is a prevalidated kit.

Suggested architecture: no custom motherboard

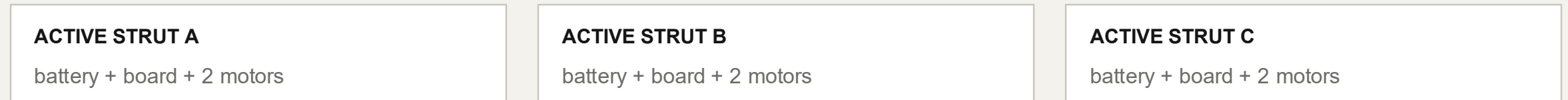
22

ONE ACTIVE STRUT = ONE COMPLETE ELECTRICAL MODULE

Suggested adaptation using Pololu #5218 + Waveshare General Driver + 3S LiPo



COPY THIS MODULE THREE TIMES



Robot hardware subtotal \$427.67 Pololu baseline	With charger + adapter + LiPo bag \$509.60 plan ~\$550–600 delivered
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No Arduino Nano · no custom motherboard · no voltage boost

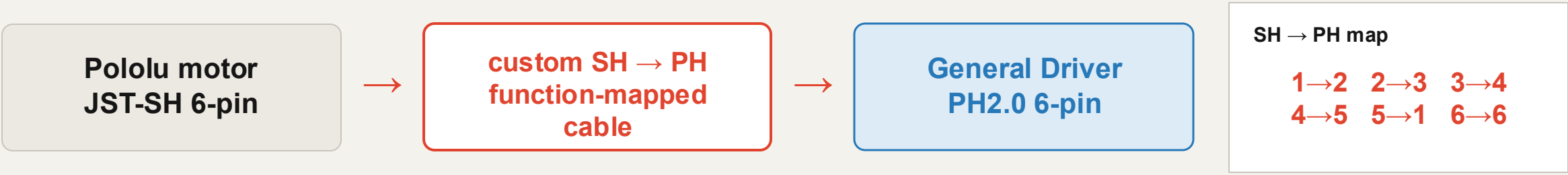
Engineering candidate: test one complete active strut before cloning.

Connectors, charging, and safety

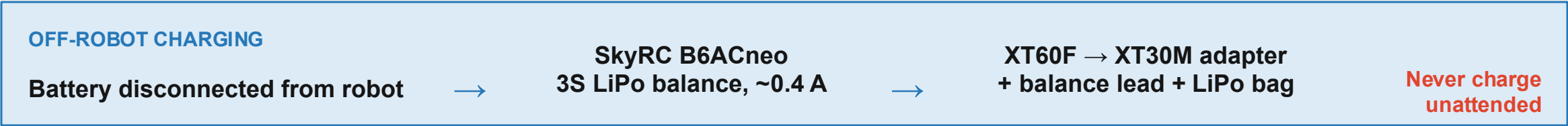
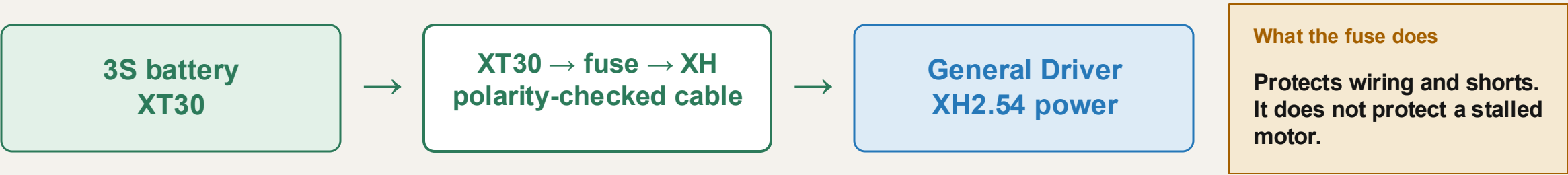
THE CHARGER IS OFF THE ROBOT; TWO CUSTOM HARNESS TYPES ARE ON THE ROBOT

Connector maps from Pololu and Waveshare documentation

MOTOR / ENCODER HARNESS · MAKE 6



BATTERY POWER HARNESS · MAKE 3



Suggested baseline BOM: Pololu build

PLANNING LIST FOR ONE THREE-BAR ROBOT — PRICES
CHECKED 7 OCT 2026

Part	Exact planning choice	Qty	Unit	Subtotal	Link
Encoder motors	Pololu #5218, 12 V, 220 rpm	6	\$29.85	\$179.10	Motor
Controllers	Waveshare General Driver, 2 motors/board	3	\$27.39	\$82.17	Board Battery
Batteries	Tattu 3S 450 mAh, XT30	3	\$12.19	\$36.57	SH PH
Motor harness parts	Pololu #4762 SH + Adafruit #5090 PH	6 sets	\$2.88	\$17.28	XT30 Fuse
Fused power harness	XT30 + XH2.54 + holder + fuse	3 sets	\$10.08	\$30.24	Tube Pre-cut
Pre-cut carbon struts	Goodwinds 0.315 in tube; supplier cuts 3	1 set	\$32.59	\$32.59	Line Cord
Active tendon line	HERCULES 70 lb / 0.44 mm PE braid	1 spool	\$11.93	\$11.93	Bearing M3
Passive elastic line	1/8 in shock cord; cut 3 tendons	1 pack	\$7.99	\$7.99	CAD
Bearings + M3 hardware	6× MR83ZZ + inserts + screw kit	1 set	\$29.80	\$29.80	Charger Adapter Bag
Printed end caps / spools	Project CAD adaptation + lab filament	1 set	\$0.00	\$0.00	
ROBOT HARDWARE			WITH ONE CHARGING SET		
\$427.67		charger not included	\$509.60		
			expect ~\$550–600 delivered		

Planning numbers exclude tax and shipping. Any substitution requires a new voltage, current, connector, and CAD check.

Students must not cut, drill, or sand carbon fiber.

Next steps before October 20

25

BOM AND CAD REVIEW

1 Choose the robot and reference path

Start with three-bar unless there is a strong reason not to. Use Yale as a published reference; select the Pololu baseline or justify the lower-cost option.

2 Prepare a BOM + wiring diagram

List the exact motor, controller, battery, connectors, charger, structure, and tendons. Include quantities, current prices, and purchase links.

3 Begin CAD around real dimensions

Design one strut, one end cap, one spool, board and battery mounts, cable guides, and strain relief. Carbon fiber must be supplied pre-cut.

OCTOBER 20 REVIEW

Bring the BOM, wiring diagram, and preliminary CAD. No purchase is required before the review.